

RECON

§11.4.153 Feature-Ledger Reconciliation — wave-7..9 planning

Revision: 1 **Last modified:** 2026-06-16T15:04:12Z **Run-id:** feature_ledger_recon_20260616T150412Z **Authority:** §11.4.153 (per-feature video ledger), §11.4.118 (code-vs-ledger reconciliation / discovery-coverage), §11.4.6 (no-guessing — every line below is a captured fact from the actual files). **Stream:** BACKGROUND operator-authorized, READ-ONLY, NON-COMMITTING. This agent did NOT edit/stage/commit/git add (a separate committer agent owns the main checkout, §11.4.84). This file is a file-write only. **Main checkout HEAD at start:** 0726c20. **Provider keys observed (§11.4.6, values never printed):** DEEPSEEK_API_KEY SET, GROQ_API_KEY SET. No other provider keys present in this shell.

1. What waves 1–6 ALREADY confirmed (the baseline — do NOT re-record as net-new)

Established from docs/features/Status.md (Rev 14) Anti-bluff note + docs/qa/video_wave*/EVIDENCE.md + docs/qa/wave3e-multipass-regression/FINDING.md. Per the doc's own §11.4.6 re-derivable count, **48 video-confirmed** rows after wave 5 (43 at Rev 14 + 5 wave-5), with wave 6 adding 5 more (HTML→HTML, PDF-input, DOCX-input, EN→ZH, EN→RU) → ~53 confirmed; the Status.md summary cell has not yet absorbed waves 4–6 (committer lag, NOT a code gap).

CONFIRMED — do not repeat:

- **Core translate (CLI):** DeepSeek EN→ES translate; -script Cyrillic↔Latin; Serbian-Cyrillic; format-detection; help/version; -verify flag.
- **Format conversions (CLI):** EPUB→TXT, HTML→EPUB, FB2→EPUB, FB2→FB2, EPUB→EPUB, TXT→TXT, HTML→HTML; HTML→Markdown; markdown EPUB↔MD round-trip.
- **Format input (CLI):** PDF input accepted, DOCX input accepted.
- **Format output (CLI):** DOCX output, PDF output (+ EPUB/FB2/TXT/HTML/MD).
- **Language pairs (CLI):** EN→FR, EN→DE, EN→IT, EN→ES, EN→PT, EN→JA, EN→ZH, EN→RU, RU→SR(cyrillic).

- **Providers (CLI per-provider path):** deepseek, cerebras, sambanova, gemini, zhipu, cohere, hyperbolic, nvidia, openrouter, fireworks, novita, mistral, groq, siliconflow.
- **REST API (live cmd/server):** POST /api/v1/translate, GET /health · /version · /providers, POST /api/v1/convert/script, POST /api/v1/translate/fb2 (→ real Serbian EPUB, hardcoded ru→sr target — honest limitation noted).
- **gRPC:** TranslationService EN→ES (CLI clip) AND **full async file-job round-trip on the LIVE nezha stack** (docs/qa/ nezha_grpc_roundtrip_20260616T143649Z/RESULT.md — StartTranslation→poll GetTranslationStatus→completed→valid translated EPUB with real Serbian; PASS).
- **Bridge / LLMsVerifier CLI:** cmd/model-bridge best-model translate + MCP-stdio surface; cmd/verify-models discovery/scoring/selection; cmd/workable-items SQLite SSoT validate+list.
- **Other binaries:** markdown-translator, preparation-translator (FIXED), cmd/cli (DeepSeek), monitor-server (live WS hub), -multipass (confirmed ONLY with explicit valid -model deepseek-chat; see §5 known bug).

2. Classification of remaining PENDING features (CLOSED set, §11.4.6)

The 186 PENDING ledger rows + Web/Library/Submodule rows were classified into the closed set the operator specified. Counts are of distinct ledger rows (flags + internal types that carry N/A are excluded from PENDING).

Classification	Count (approx, distinct PENDING rows)	Meaning
[CLI-LOCAL-CONFIRMABLE]	~38	exercisable via ./build/unified-translator or local ./build/server with the keys present (deepseek/groq), no live nezha/LLMsVerifier server needed
[LIVE-BRIDGE-NEEDED]	~30	needs a reachable LLMsVerifier server (localhost:8080) and/or the additional provider keys that are EMPTY in the env — blocked by the operator-review item; DEFER
[WEB-DASHBOARD]	~37	needs a browser/HelixQA driving monitor.html / enhanced-monitor.html / dashboard.html — operator-attended
[GRPC]	0 net-new	the 7 RPCs class is already LIVE-confirmed via the nezha async round-trip; remaining per-RPC rows are sub-features of that confirmed flow

Classification	Count (approx, distinct PENDING rows)	Meaning
[DESKTOP/ MOBILE]	0	no desktop/mobile surface exists in this codebase
[NON-FEATURE / N/A]	258 (the N/A rows)	bare CLI flags, internal library types, infra middleware — no standalone user-visible video applies

LIVE-BRIDGE-NEEDED detail (DEFER, §11.4.6): the `-use-verifier` path requires a live LLMsVerifier at `http://localhost:8080` which is unreachable in this environment (every wave EVIDENCE.md records this honest constraint). Additionally, per `docs/qa/operator_review_20260616T142856Z/PLANS.md`, 9 provider keys are EMPTY in the nezha container (OPENAI, ANTHROPIC, QWEN, XAI, TOGETHER, NLP_CLOUD, SARVAM, LLMSVERIFIER, SSH_WORKER_PASSWORD) — so OpenAI/Anthropic/Qwen/xAI/Together/NLPcloud/Sarvam providers CANNOT be confirmed until the operator populates those keys. These belong to LIVE-BRIDGE-NEEDED, not CLI-LOCAL.

WEB-DASHBOARD detail: `monitor.html` (9 rows), `enhanced-monitor.html` (8 rows), `dashboard.html` (7 rows), `monitor-server /ws,/monitor,/health` (4 rows) + `version-monitoring dashboard.html`. All require browser/HelixQA per §11.4.143/§11.4.117 — operator-attended; not autonomously CLI-confirmable.

3. CLI-LOCAL-CONFIRMABLE inventory (the actionable pool for waves 7–9)

These ARE exercisable now with `DEEPSEEK_API_KEY/GROQ_API_KEY` and `./build/unified-translator / ./build/server`. Distinct from waves 1–6.

A. Native-factory providers via `unified-translator -provider <name>` (key permitting). `pkg/translator/llm/llm.go` switches 26 native providers directly. Confirmed already: `deepseek`, `cerebras`, `sambanova`, `gemini`, `zhipu`, `cohere`, `hyperbolic`, `novita`, `mistral`, `groq`, `siliconflow` (+ `nvidia/openrouter/fireworks` via submodule path). **Remaining native providers with code BUT no key in this env** → these are CLI-shaped but key-blocked, so effectively LIVE-BRIDGE/key-blocked: `openai`, `anthropic`, `qwen`, `xai`, `togetherai`, `nlpcloud`, `sarvam`, `upstage` (403). **Remaining native providers possibly key-available depends on env:** `kimim`, `modal`, `publicai`, `nia`, `vulavula`, `replicate`, `cloudflare` — confirm `<NAME>_API_KEY` presence before recording; only those with a real key are genuine CLI-LOCAL.

B. NOT-yet-confirmed CLI-local FEATURES that work on `deepseek/groq` (no extra key):

- GROQ provider** EN→ES via `-provider groq` (key is SET; only `deepseek`-family confirmed for many feature rows) — a second-provider liveness confirmation distinct from the `deepseek` baseline.
- More language pairs** through the confirmed `deepseek` path that the matrix has not yet covered, e.g. **EN→SR(latin)**, **EN→AR**, **EN→KO**, **DE→EN** (reverse

- direction), **FR→ES** (non-English source→non-English target) — each a distinct user-visible capability the `-source-lang/-target-lang` rows enable (Status.md lines 271–272 are N/A as bare flags; the CAPABILITY deserves §11.4.153 confirmation per the wave-4 precedent).
3. **TXT→EPUB / TXT→DOCX / TXT→PDF / TXT→FB2** output-format cross-products from a plain TXT source (writers all confirmed present in code; the specific source→target cross-products are net-new rows).
 4. **EPUB→DOCX / EPUB→PDF / EPUB→HTML / EPUB→MD** conversions (EPUB parser + each writer confirmed in code; these specific conversions not yet recorded).
 5. **-chunk-size / -concurrency / -workers** real effect on a multi-section book (a session-report-backed confirmation that chunking actually parallelises — distinct from bare-flag N/A).
 6. **-temperature / -max-tokens** producing a genuinely different output on the same input (liveness via output-diff, deepseek).
 7. **REST POST /api/v1/translate/string** and **POST /api/v1/translate/directory** (pkg/api/batch_handlers.go) against the local cmd/server — distinct from the confirmed /ttranslate JSON endpoint.
 8. **REST POST /api/translate + POST /api/upload + GET /api/languages** on the standalone pkg/api/server.go gin server (a different server impl than cmd/server's handler.go).
 9. **cmd/preparation-translator analysis JSON** consumed by a follow-on translate (the two-stage pipeline end-to-end, deepseek) — preparation row is confirmed in isolation but the consume-the-analysis flow is net-new.
 10. **cmd/markdown-translator MD→EPUB direction** specifically (round-trip confirmed; the MD-as-source→EPUB direction as a standalone is distinct).
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4. RECOMMENDED waves 7–9 (CLI-LOCAL, no overlap with waves 1–6, deepseek/groq only)

Ordered by value × distinctness; each batch = 3–5 net-new, no cross-batch overlap. All confirmable with the present keys + local binaries.

Wave 7 — output-format cross-products (highest distinctness, all writers code-confirmed)

1. **TXT→EPUB** (EN .txt → real Spanish EPUB, open artifact + read chapter)
2. **TXT→PDF** (EN .txt → valid Spanish PDF)
3. **EPUB→DOCX** (EN .epub → real Spanish .docx, open OOXML)
4. **EPUB→HTML** (EN .epub → Spanish .html)
5. **TXT→FB2** (EN .txt → Spanish FB2 with namespace preserved)

Wave 8 — language-matrix breadth + second provider (capability rows the flags enable)

1. **GROQ provider EN→ES** (second live provider, key SET — distinct from all deepseek rows)
2. **EN→SR(latin)** translate (latin Serbian, distinct from confirmed cyrillic)
3. **EN→KO (Korean)** non-Latin output (distinct from JA/ZH)
4. **DE→EN** reverse-direction translate (source≠English)

5. **FR→ES** non-English-source→non-English-target (distinct pair class)

Wave 9 — REST surface breadth + pipeline depth (local cmd/server / pkg/api/server.go)

1. **REST POST /api/v1/translate/string** real JSON translated string (batch_handlers.go)
2. **REST POST /api/v1/translate/directory** real directory batch (batch_handlers.go)
3. **REST POST /api/translate** on standalone pkg/api/server.go gin server (distinct impl)
4. **REST POST /api/upload** file-upload→translate on pkg/api/server.go
5. **preparation→translate two-stage pipeline** (preparation-translator analysis JSON consumed by unified-translator, deepseek)

§11.4.6 caveat on Wave 8.1/native providers: only record a provider whose <NAME>_API_KEY is actually present at record time. groq is SET. For any other provider, verify the key first or it is an honest SKIP, never a faked PASS (the wave-3d upstage-403 / together-no-key precedent).

5. §11.4.118 code-vs-ledger reconciliation findings (mismatches / honest notes)

Verified against actual code (pkg/translator/llm/*.go, llm_provider/pkg/providers/*, pkg/ebook/*, pkg/grpc/server.go, pkg/api/verifier_handlers.go, cmd/).

- **[MATCH]** All 26 native factory providers in pkg/translator/llm/llm.go (lines 264–315) have real client .go files. No claimed native provider lacks code.
- **[MATCH]** All 7 output writers present: epub_writer.go, fb2_writer.go, docx_writer.go, pdf_writer.go; TXT/HTML/MD via cmd/unified-translator/main.go output switch. All 6 input parsers present: fb2/epub/docx/html/pdf/txt_parser.go.
- **[MATCH]** All 7 gRPC RPC handlers present in pkg/grpc/server.go.
- **[MATCH]** GET /api/v1/verified-models handler EXISTS (pkg/api/verifier_handlers.go:100/109); the runtime 404 is config-driven (LLMsVerifier disabled), NOT missing code — Status.md states this honestly.
- **[MATCH]** All 4 Obsolete cmd dirs (translator, translate-ssh, ebook-translator, ssh-translation) are genuinely ABSENT from disk — Obsolete classification is accurate.
- **[MISMATCH-no-ledger / minor]** cmd/grpc-translate-probe EXISTS on disk (used as the nezha gRPC wire-client in docs/qa/nezha_grpc_roundtrip_*) but is NOT enumerated as a section in docs/features/Status.md. It is a test/probe client, not a shipped user binary, so the omission is minor — recommend adding a one-line ledger row classified N/A (internal probe) or Confirmed (it IS the tool that produced the gRPC round-trip evidence). Honest §11.4.118 gap, not a code-absence.
- **[NOTE]** The llm_provider submodule carries ~44 provider dirs; only the subset reachable through the unified-translator/bridge path is user-exercisable. The extra submodule providers (ai21, chutes, codestral, githubmodels,

huggingface, junie, kilo, ollama, perplexity, venice, zai, zen, generic, claude, ai21, ...) are NOT enumerated as confirmable HelixTranslate features and are LIVE-BRIDGE-only — correctly out of the CLI-LOCAL pool.

- **[NOTE — real bug, already tracked]** `-multipass` default-model path is BROKEN (docs/qa/wave3e-multipass-regression/FINDING.md + wave-5 EVIDENCE): default `gpt-4` is rejected by the `ValidModels` whitelist for deepseek; the polish path also has no guard against the LLM returning a meta-response (writes garbage as the translation). Confirmed working ONLY with explicit `-model deepseek-chat`. The ledger correctly demotes it to `PENDING_FORENSICS`. Do NOT re-record multipass as a clean confirmation until the source fix lands.
 - **[NOTE — real bug, already tracked]** `POST /api/v1/translate/fb2` hardcodes `ru→sr` (pkg/api/handler.go:84–85), ignoring requested target. Feature confirmed working, hardcoded-target is an honest limitation.
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6. Counts summary (for the report)

- **Video-confirmed baseline (waves 1–6):** ~53 distinct rows (Status.md re-derivable 48 post-wave-5 + 5 wave-6; summary cell lags at 43 — committer catch-up owed, not a code gap).
- **PENDING classification:** CLI-LOCAL-CONFIRMABLE ~38 · LIVE-BRIDGE-NEEDED ~30 · WEB-DASHBOARD ~37 · GRPC 0 net-new (class already live-confirmed) · DESKTOP/MOBILE 0 · NON-FEATURE/N/A 258.
- **Code-vs-ledger mismatches:** 1 minor ledger gap (`cmd/grpc-translate-probe` unlisted); 0 claim-without-code. Ledger is materially accurate.
- **Recommended next 3 waves:** 15 CLI-local net-new confirmations (Wave 7 output cross-products, Wave 8 language-matrix + groq, Wave 9 REST breadth + pipeline depth) — all autonomously recordable with deepseek/groq, no live nezha bridge required.